A Minor Project Synopsis

On

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Signature of Supervisor: _____

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Introduction

The Restaurant Booking System is a web-based application designed to simplify table reservations in a restaurant.

It allows users to book tables by entering details such as name, date, and time.

The system helps in reducing manual effort and prevents double booking by checking slot availability.

It is developed using Python with Flask, along with HTML and CSS for the user interface.

Problem Statement

In many restaurants, table reservations are still managed manually, which can lead to inefficiencies such as double booking, long waiting times, and poor customer experience. Customers often face inconvenience in reserving tables in advance, especially during peak hours.

There is a need for a simple and efficient system that allows users to book tables easily and ensures proper management of reservations. The system should be able to check availability and prevent conflicts in booking.

Objectives

- To develop a web-based Restaurant Booking System
- To enable users to book tables by selecting date and time
- To prevent double booking of tables
- To provide a simple and user-friendly interface
- To improve efficiency in managing reservations

Methodology

The project follows a structured approach to develop the Restaurant Booking System.

- First, the user enters booking details such as name, date, and time through a web form created using HTML and styled with CSS.
- The form data is sent to the backend using Flask.

- The backend processes the input and checks whether the selected date and time slot already exists in the booking records.
- If the slot is already booked, the system displays a message indicating that the slot is not available.
- Otherwise, the booking is confirmed and stored in a list.

The system is tested by performing multiple booking operations to ensure that duplicate bookings are prevented and valid bookings are successfully recorded.

📜 Tools and Techniques

➢ Tools Used

- Python – used for implementing the core logic of the system
- Flask – used as a backend framework to handle requests and responses
- HTML – used to design the structure of the web pages
- CSS – used for styling and improving the user interface
- Visual Studio Code – used for writing and executing the code
- Bootstrap - a CSS framework to improve the design and responsiveness of the user interface

➢ Techniques Used

- Client-Server Architecture – interaction between frontend and backend
- Form Handling – collecting user input through web forms
- Conditional Logic – checking availability and preventing double booking

- Data Storage using Lists – temporary storage of booking data

This project demonstrates the application of software engineering principles to solve a real-world problem of managing restaurant reservations.

Expected Outcomes

- A functional web-based Restaurant Booking System
- Users can successfully book tables by entering date and time
- The system prevents double booking of the same slot
- Displays confirmation or error messages based on availability
- Provides a simple and user-friendly interface for reservations

Applications

- Used in restaurants to manage table reservations efficiently
- Helps customers book tables in advance without visiting physically
- Reduces waiting time and avoids overcrowding
- Can be used in hotels, cafes, and event venues for booking management
- Improves overall customer experience and service efficiency

Conclusion

The Restaurant Booking System successfully demonstrates a simple and efficient way to manage table reservations. It allows users to book tables easily and prevents double booking through proper validation.

The project highlights how basic web technologies like Python with Flask, HTML, and CSS can be used to build a functional real-world application. Although the system has some limitations, it can be further improved by adding advanced features and better design.